A Unified Study of Influence Maximization in Social Networks: HIM, DeepSN, and GBIM

Course Project Milestone Report
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Abstract

This milestone defines a unified course project on social-network information diffusion and influence maximization. Three members prepare complementary methods: HIM learns hyperbolic user representations from a graph and propagation histories; DeepSN uses a sheaf-neural surrogate for diffusion-aware seed selection; and GBIM applies Bayesian optimization to multiplex user–item influence. HIM and DeepSN both have verified team environments and end-to-end experiments. On Cora-ML, one fixed-seed HIM WIC+ASW run produced 37.6195% independently evaluated coverage at its saved final epoch. A same-checkpoint ASW/MD pairing on that Cora-ML model found a small ASW advantage whose 95% interval excludes zero; the Ciao three-seed ASW/MD interval includes zero. DeepSN reproduced the authors' Jazz SIS protocol at 86.4495% coverage and completed a Ciao three-seed WIC study aligned with HIM's graph, probabilities, budget, and evaluation stream (mean coverage 8.132%). GBIM has a preliminary native Multi-IC reproduction on author-provided data; its Ciao item-aware adapter has passed schema and small-budget smoke checks but not a full multi-seed evaluation. Neither Ciao three-seed interval supports a between-method ranking. Ciao is the primary new-data target because it combines user trust with user–item interactions. Direct three-way ranking will be reported only if a GBIM adaptation passes an explicit objective-equivalence check.

1 Problem and Project Scope

We select the project theme of social-network transmission analysis, with influence maximization as the central task. Given a graph $G = (V, E)$, a diffusion process $\mathcal{M}$, and a budget k, the goal is to choose an initial seed set

$$S^* \in \arg\max_{S \subseteq V,\ |S| \le k} \sigma_{\mathcal{M}}(S), \qquad \sigma_{\mathcal{M}}(S) = \mathbb{E}\big[|A_{\mathcal{M}}(S)|\big], \tag{1}$$

where $A_{\mathcal{M}}(S)$ is the set eventually activated from S. Diffusion rules, edge probabilities, observed histories, and even the meaning of a "seed" differ across methods. Two individually strong users may reach much of the same audience, so high degree alone does not solve the set problem.

Our overall question is: *How do graph-based influence methods use structural, propagation, and user–item evidence to select effective interventions under a limited budget?* The primary dataset is the Ciao review/trust network. The methods share one data pipeline and a staged experiment: first compare user-level seed selection under one fixed propagation evaluator; then test whether item-aware user–item actions improve reach on the same users and trust graph. DeepSN's native author protocol uses SIS on Jazz; the shared Ciao contract uses weighted independent cascade (WIC). Those numbers answer different questions and must not be pooled. The intended contribution is a reproducible comparison pipeline, a careful statement of information assumptions, and a small empirical study rather than a claim of universal superiority.

2 Related Methods and Their Connection

Classical influence maximization maximizes expected spread under an explicit diffusion oracle. Under standard IC/LT assumptions the objective is monotone and submodular, so greedy selection obtains a $1 - 1/e$ approximation given an exact marginal-gain oracle [1]. A discrete SIS-style contact process allows recovered nodes to become susceptible again, so a reported infected share need not correspond to an absorbing activated set [2]. The assigned methods move beyond a single greedy oracle:

- **HIM** learns one user embedding from social edges and observed activations. Lorentzian distance to the origin estimates individual influence; adaptive sliding window (ASW) raises scores of nearby candidates to reduce likely overlap [3].
- **DeepSN** learns a map from a seed indicator to an activation pattern. Cellular-sheaf restriction maps allow heterogeneous neighbor-to-neighbor influence [5, 6, 4]. Seed assignment then uses community-proportional budget allocation rather than a single radial score.
- **GBIM** uses graph Bayesian optimization for multiplex influence maximization. Its native decision is over user–information pairs, not simply a set of users [7].

DeepIM is a related seed-set representation method, but is not one of the three assigned implementations [10]. The

methods are complementary views of one pipeline—graph and interaction data; representation or surrogate learning; seed or action selection; independent evaluation—and should not be ensembled merely because they share the phrase "influence maximization."

3 Data Collection and Initial Findings

3.1 HIM: Cora-ML and Email-Eu-core

The verified HIM reproduction uses the author-provided Cora-ML graph with 2,810 nodes and 7,981 undirected edges. Its WIC training cache contains 30 propagation instances. The saved run uses WIC, ASW, seed ratio 10% ($k = 281$), training seed 42, and stops at epoch 54. The saved final-epoch seed set was independently evaluated for 1,000 simulations with evaluation seed 1042: mean activated users 1,057.107, standard deviation 32.1214, coverage 37.6195%. Reloading the checkpoint reproduced that seed set and summary. A distinct maximum *validation* coverage of 38.15% occurred at epoch 43; it must not be reported as the independent score of the saved epoch-54 model. A later same-checkpoint ASW/MD pairing (also 1,000 rounds, seed base 1042) gave mean activated 1,057.466 vs. 1,054.06 and a paired difference whose 95% interval excludes zero. That pairing is a selector ablation on one Cora-ML checkpoint, not a new independent last-epoch score, and not a Ciao result.

The paper specifies WIC probabilities in $[0, 0.2]$, but the bundled Cora-ML probability file spans approximately $[0.1, 0.2]$. The paper describes WLT parameters as fixed across realizations, whereas the repository resamples node thresholds in each WLT simulation. The verified run therefore reproduces the recorded code/data setup, not every manuscript setting. The author's Cora-ML WIC table (about 38.87% for HIM at 10% budget) is a different training/evaluation contract and is not our 37.6195% figure [3].

An exploratory Email-Eu-core Department 3 sample from SNAP covers 89 users and 3,514 events in a 180-day window [9]. Retaining directed edges observed at least twice gives 464 edges, 15 isolates, and 16 weakly connected components. It is a parser smoke test, not evidence of real opinion adoption.

3.2 DeepSN: native Jazz

DeepSN's author protocol is tied to the Jazz collaboration network in the public repository (commit `bbc7780`), not to Ciao. The local cache `jazz_mean_SIS200.SG` (SHA-256 prefix `8e7f1eed`) has 198 nodes and 100 seed–activation pairs under SIS and is not committed to Git. On an Apple Silicon Mac Air (Python 3.11.17, PyTorch 2.5.1 CPU), one native run with 20% budget ($k = \lceil 0.2 \cdot 198 \rceil = 40$), training seed 42, 50 estimation plus 200 scoring epochs, and 100 SIS evaluations produced mean coverage 86.4495% (sd 2.173 percentage points; mean activated 171.17). That figure describes a dense 198-node contact process. It is not a Ciao result and is not comparable to HIM's Cora-ML WIC score.

3.3 Primary dataset: Ciao

Ciao is the primary new dataset because it provides a trust graph and user–item review relations. We retrieved the baseline MAT archive from the dataset authors' page [8]. The page states that the collections were crawled in May 2011 and that the published files are a partial release. The archive contains 284,086 ratings and 111,781 directed trust arcs. The trust graph covers 7,317 users; its undirected projection has 85,205 edges, seven connected components, and a largest component of 7,305 users. Rating records cover 7,375 users, of whom 58 have no trust-network membership (766 rating rows); the shared graph-aligned rating table therefore retains $284{,}086 - 766 = 283{,}320$ source rows. The Ciao page does not state an explicit license. Raw MAT files stay local, are not redistributed, and are cited; course-use expectations will be confirmed before any public release. Provenance is in `data/ciao_source_manifest.json`.

The baseline MAT archive has no rating timestamps, and trust links have no establishment dates. A separately packaged timestamped text release has 57,544 trust rows and 36,065 timestamped rating rows, substantially fewer than the MAT archive. It is excluded rather than merged. Shared preprocessing creates sorted contiguous user/item maps, preserves directed trust edges, exports a deduplicated undirected user graph, and retains in-scope rating rows (including duplicate pairs and zero ratings). GBIM's adapter instead keeps the first rating per user–item pair, removes zero ratings, filters users outside the trust graph, and symmetrizes trust edges, yielding 281,865 ratings over 7,317 users and 104,957 items.

Ratings are product feedback, not evidence that one user caused another's adoption. User-level experiments therefore use a documented synthetic WIC generator on this topology, with training and evaluation banks kept separate. The shared rule is

$$p(u, v) = p(v, u) = \frac{1}{\max(\deg u, \deg v)}, \tag{2}$$

recorded as the shared Ciao WIC `v1` protocol. Budget $k = 366$ is the nearest integer to 5% of 7,317 (DeepSN's `ceil` rule yields the same k). HIM and DeepSN each draw 30 training cascades per seed $s \in \{42, 43, 44\}$; they do *not* share cache files. Independent evaluation uses 1,000 WIC rounds and seed base 1042.

3.4 Ciao summary statistics

Table 2 lists verified scores. HIM-ASW three-seed mean activated users is 735.03 (coverage 10.046%); the paired

Table 1: Method roles, inputs, and comparison boundary.

Method	Native decision	Main evidence	Fair-comparison requirement
HIM	k user nodes	Social graph and propagation instances	Fix graph, cascade bank, budget, and independent evaluator; compare ASW with HIM-MD.
DeepSN	k user nodes / learned seed set	Graph and generated or observed diffusion examples	Use the real CLI; set sheaf `graph_size` to $\|V\|$; keep training cascades off the evaluator stream; report SIS/Jazz and WIC/Ciao as distinct protocols.
GBIM	User–item seed pairs	Social and user–item/item relations	Native Multi-IC pipeline preliminarily reproduced; adapt to a shared user-only task only after validating the changed objective and code.

ASW–MD difference interval includes zero. DeepSN three-seed mean activated users is 595.05 (sd 20.30; coverage 8.132%; 95% t interval $[544.63, 645.47]$, df= 2). Seed 44 (571.62) is well below seeds 42 and 43 (606.47 and 607.06). Trainers differ (DeepSN: 50+200 epochs, no HIM-style 100-round online validation), so the gap is not a ranking.

A native DeepSN IC evaluator ($p = 1$/in-degree) on an earlier Ciao *lite* run (16 training pairs, 5+20 epochs, 20 IC rounds) reported 26.09% coverage. That is not the three-seed n=30 seed sets, which were saved with `eval_rounds=0` and scored only under independent WIC. Switching the evaluator changes the task; the lite IC number must not be attached to the formal seeds.

4 Mathematical Background and Algorithms

In IC, each newly active node gets one attempt to activate each inactive neighbor with probability p_{uv}. In LT, node v activates once incoming active-neighbor weight reaches θ_v. A discrete SIS process allows return to susceptibility, so the reported quantity is an infected share rather than a terminal set [2]. These models permit controlled evaluation; they do not prove that Ciao users follow them. Expected spread is expensive, which motivates simulation or learned surrogates.

HIM uses Lorentz hyperbolic space, where negative curvature can represent hierarchy compactly [11]. It fits social edges and activation edges with relation-specific rotations and negative sampling. A proactive term moves users observed to activate others closer to the Lorentz origin. The strength proxy is

$$\mathrm{LDO}_u = d^2_{\mathcal{L}}(\mathbf{x}_u, \mathbf{o}_{\mathcal{L}}), \tag{3}$$

with smaller LDO implying higher estimated influence. ASW ranks by ascending LDO in a window of size $w \approx \beta k$. After selecting c, it increases scores of candidate neighbors by hyperbolic proximity to c. This approximates audience overlap locally; no greedy $1 - 1/e$ guarantee follows.

DeepSN replaces repeated marginal-gain oracles with a two-stage learned procedure [4]. A cellular sheaf assigns a stalk $\mathbb{R}^d$ to each node and a restriction map $F_{u\to v}$ to each oriented edge; Neural Sheaf Diffusion parameterizes the sheaf Laplacian L_F from node features [5, 6]. Stage 1 maps local structural one-hots through a sheaf block and learnable sigmoid edge weights to a surrogate $\hat{a} = f_\theta(s)$. Stage 2 freezes f_θ, partitions the learned-weight graph with Louvain, and trains a shallow scorer; budget k is split by community size, with remainder filled by global high scores. The 0/1 assignment is hard, so gradients are weak. The course CLI now reads arguments, no longer lets sheaf parsing steal `-d`, and sets `graph_size` to current $|V|$ instead of Jazz's 198. Author training and selection loops are otherwise unchanged. Ciao training histories are WIC-generated even when the CLI flag is `IC`; independent scores use the shared WIC evaluator.

GBIM treats multiplex IM as black-box combinatorial optimization over user–information pairs, with a Multi-IC simulator and a graph-aware Bayesian surrogate. That action space is larger than a user-only set. The shared Ciao user-only benchmark can include GBIM only after constraining the action space and checking the objective. In this milestone the author LastFM/Delicious pipeline has been run end-to-end as a preliminary single-configuration reproduction; Ciao item-aware smoke checks pass; full multi-seed Ciao runs remain.

The design separates training information from evaluation information. Greedy/CELF, if included, must be marked as receiving true simulator parameters when learned methods see only generated histories. Degree and random selection are inexpensive controls. The primary metric is $100\,\mathbb{E}[|A(S)|]/|V|$, with intervals across training seeds, plus wall-clock for preparation, training, selection, and evaluation.

5 Current Progress, Difficulties, and Work Plan

Main risks: (i) the methods do not optimize identical objects; (ii) reproduction depends on diffusion parameters and whether histories are shared; (iii) preprocessing can change direction, identity, or k; (iv) training randomness and finite Monte Carlo add uncertainty; (v) ratings/trust are not transmission logs; (vi) Ciao has no explicit license. DeepSN already hit three implementation risks: the original CLI did not switch the default SIS simulator; sheaf parsing collided with dataset `-d`; and `graph_size`=198 would have made Ciao wrong. Torch 2.14 crashed against published PyG `torch-sparse` wheels on Apple Silicon,

Table 2: Verified user-level scores in `results/`. Jazz SIS and Ciao WIC are different tasks.

Protocol	Method	k	Mean $\lvert A(S)\rvert$	Cov. %	Note
Cora-ML WIC, seed 42	HIM-ASW, epoch 54	281	1057.11	37.6195	1,000 rounds, eval seed 1042
Cora-ML WIC, same checkpoint	HIM ASW vs. MD	281	+3.41	—	Paired 95% CI excludes 0; not Ciao
Jazz SIS, seed 42	DeepSN	40	171.17	86.4495	Author SIS; 100 rounds
Ciao WIC, seeds 42/43/44	HIM-ASW	366	735.03	10.046	30 cascades/seed; ASW–MD CI includes 0
Ciao WIC, seeds 42/43/44	DeepSN	366	595.05	8.132	30 cascades/seed; t interval df= 2

Table 3: Team status at this milestone.

Workstream	Evidence and remaining work
HIM	Cora-ML independent evaluation and same-checkpoint ASW/MD ablation verified; Ciao three 30-instance seeds plus 1,000-round held-out ASW/MD complete. Across-seed paired interval includes zero; add degree/random controls.
DeepSN	Mac Air CPU verified. Jazz SIS complete (86.45%). Ciao matched WIC three-seed independent evaluation complete (8.132%). Next: degree/random controls and more training seeds.
GBIM	Preliminary native Multi-IC reproduction (single configuration). Ciao item-aware adapter: schema/small-budget smoke. Next: fixed-seed native repeats, then user-only equivalence check.
Team	Ciao schema audited; shared preprocessing in place. Freeze the evaluator before any three-way ranking.

hence the Torch 2.5.1 lock.

Proposed next steps. Confirm educational use of Ciao; keep the shared export. Add Degree/Random on the same WIC evaluator for HIM and DeepSN; expand DeepSN from three to five training seeds so seed 44 does not dominate the interval. Repeat GBIM native Multi-IC with fixed seeds; constrain to user-only actions only if the objective still matches. Individual reviews: HIM English (Liu Shengqi) and DeepSN Chinese (Li Zixi) are written; GBIM review is in progress as a separate document. The final-project date remains to be confirmed.

6 Conclusion

The project is feasible around one Ciao source and a staged comparison of complementary IM paradigms. HIM has a verified Cora-ML result and a same-checkpoint selector ablation; Ciao three-seed paired evaluations do not establish an ASW advantage or a cross-method ranking. DeepSN has a verified Jazz SIS result and a Ciao three-seed WIC evaluation that shares HIM's graph, probability identifier, $k = 366$, and evaluation seed 1042 (8.132% vs. HIM-ASW 10.046%). That juxtaposition is a protocol match, not a ranking. GBIM has a preliminary native reproduction and a Ciao smoke check, not full multi-seed Ciao checks. Next: confirm data-use expectations, add degree/random controls, complete GBIM's native protocol, and freeze the user-level evaluation contract.

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